

David Osorio

832-487-6377 | osoriodav2@gmail.com | osoriodav2.github.io

EDUCATION

Wesleyan University

Middletown, CT

Bachelor of Arts in Computer Science and Mathematics (GPA: 3.71 / 4.0)

May 2028

- **Relevant Coursework:** Discrete Mathematics, Applied Data Analysis, Computer Science I/II

Houston City College

Houston, TX

Associate of Science in Computer Science (GPA: 3.97 / 4.0)

May 2025

- Graduated with Highest Honors | Honors College Graduate
- **Relevant Coursework:** Data Structures and Algorithms, Computer Organization and Architecture

RESEARCH EXPERIENCE

NSF Undergraduate Researcher

Houston, TX

University of Houston-Downtown

June 2026 – Present

- Architected privacy-preserving Python data pipelines utilizing native NumPy and pandas to process complex UCI machine learning datasets, quantifying the tradeoff between Local Differential Privacy (LDP) mechanisms including Direct, Symmetric, and Optimal Unary Encoding and baseline Naive Bayes classification utility.
- Engineered custom dimensionality reduction (PCA/DCA) algorithms and equal-width discretization frameworks to benchmark statistical accuracy retention and data security across continuous clinical datasets under dynamic privacy budgets.

Student Researcher

Middletown, CT

Wesleyan University

January 2026

- Spearheaded quantitative climate change research on a national U.S. survey ($N = 929$), analyzing demographic and regional variables to identify key statistical predictors.
- Executed advanced bivariate and multivariate statistical analyses including Chi-square, ANOVA, and logistic/linear regressions using R to extract actionable data insights.

PROJECTS

NYC Crime Explorer | Python, pandas, SciPy, Frontend Web Maps

April 2026

- Developed an interactive data visualization web application merging NYPD criminal complaint records with U.S. Census API data to geographically map offense severity.
- Engineered dynamic frontend filters and a Python backend utilizing pandas and SciPy to execute real-time regression analyses controlling for key demographic variables.

Stock Prediction Agent | Python, Yahoo Finance API, pandas, Matplotlib

March 2026

- Designed an automated quantitative trading script in Python utilizing the Yahoo Finance API and a comprehensive backtesting framework to simulate historical market data models and forecast equity price movements.
- Engineered a real-time analytical module for automated equity trading, generating actionable insights and optimizing risk management based on historical volatility metrics.

WORK EXPERIENCE

Service Desk Student Worker

Middletown, CT

Wesleyan University

September 2025 – Present

- Delivered Tier 1 IT support, independently resolving critical identity and access management incidents across Duo Security, Microsoft Outlook, and Google Workspace.
- Diagnosed and resolved diverse hardware and software malfunctions for students and faculty, ensuring continuous uptime for campus network systems and printing devices.

IT Technician

Houston, TX

University of Houston at Katy

October 2024 – August 2025

- Maintained and troubleshooted complex hardware, software, and audio-visual systems across high-volume university lecture halls to facilitate seamless classroom instruction.
- Administered critical campus IT infrastructure, optimizing network connectivity and actively troubleshooting systems for library computing and printing networks.

HONORS & AWARDS

Citation of Citizenship in the Department of Mathematics and Computer Science

May 2026

ASA DataFest CT 2026 Category Winner

Apr. 2026

Ronald E. McNair Scholar

Nov. 2025

Phi Theta Kappa Membership

Feb. 2025

Houston City College Honors College Entry/Graduation

July 2024

AP Scholar with Distinction

July 2024

Texas First Early High School Graduation Scholar

May 2024

TECHNICAL SKILLS & ADDITIONAL

Technical Skills: R, RStudio, C, C++, Python, Windows Server, Network Management, Access Control Lists

Certifications & Training: CompTIA Network+ CE, Rice University Environmental Data Academy

Languages: English (Fluent), Spanish (Fluent)